

A balancing-selected chromosome-2 supergene and a neural and photoreceptive program of recent selection in the northern shortfin squid *Illex illecebrosus*

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Abstract

Marine species with high dispersal capabilities often exhibit minimal geographic population structure while maintaining significant adaptive genomic variation. In this study, we employ a chromosome-level reference genome and whole-genome data from 633 northern shortfin squid, *Illex illecebrosus*, collected across the Northwest Atlantic, to elucidate genomic variation previously inaccessible with a divergent reference assembly. Our findings confirm nearly complete range-wide panmixia, yet reveal a singular major axis of genomic structure: a ~19 Mb inversion on chromosome 2. This derived arrangement segregates at a frequency of 0.374, is present throughout the species' range without a geographic cline, and achieves $F_{ST} = 0.37$ between alternative homozygotes despite a geographic $F_{ST} \approx 0$. Its evolutionary trajectory is inconsistent with neutral drift. The rarer derived arrangement retains 2.3 times more diversity than expected given its current frequency, and barrier-aware coalescent modeling dates the polymorphism to approximately 850 ka, indicating a prolonged period near frequency $p \approx 0.74$ followed by a decline over the last ~50–100 ky. The inversion is gene-poor and lacks functional enrichment, yet contains thousands of near-fixed differences concentrated within the non-recombining block, suggesting selection on the arrangement as a long-lived supergene rather than on a single locally adaptive locus. Outside the inversion, demographic analyses indicate strong expansion (genome-wide Tajima's $D \approx -2$) to a current effective population size of approximately seven million. A stratified scan, corroborated across methods, for recent positive selection identified 627 candidate sweep windows, including 24 high-confidence regions independently supported by multiple methods. Genes associated with these sweeps show significant enrichment for neurodevelopment and photoreception, with additional candidates involving lineage-specific genes and Z-linked RNA-regulatory and hormone-signaling loci. Collectively, these findings reveal two contrasting modes of adaptation in an otherwise panmictic marine species: a long-lived chromosomal polymorphism maintained by balancing selection and recent directional selection on sensory and neural functions. More broadly, they illustrate how a conspecific reference genome can uncover structural and adaptive variation that remains largely undetectable through geographic differentiation alone.

1 Introduction

Marine populations characterized by vast census sizes and high dispersal rates pose a persistent challenge for population genetics, as weak geographic differentiation can coexist with significant

evolutionary variation that remains undetected by conventional population structure analyses. This issue is particularly pronounced in pelagic cephalopods. Ommastrephid squids, which are short-lived, semelparous, highly fecund, and capable of extensive dispersal during both early life stages and adulthood, generate large and often highly variable populations whose abundance is closely linked to oceanographic conditions (O’Dor and Dawe, 1998; Arkhipkin et al., 2015). The northern shortfin squid, *Illex illecebrosus*, exemplifies this life history. It is an annual semelparous broadcast spawner distributed over more than 2,500 km of the Northwest Atlantic shelf and slope, from the Mid-Atlantic Bight to Newfoundland and Labrador, where it supports major commercial fisheries in both U.S. and Canadian waters (O’Dor and Dawe, 1998; Arkhipkin et al., 2015; Baker et al., 2025). Recruitment and distribution are strongly influenced by transport in the Gulf Stream and associated shelf and slope circulation, while adults undertake extensive horizontal and diel vertical migrations. Collectively, high dispersal, extreme fecundity, and large population size suggest minimal persistent geographic structure, despite providing a substantial substrate for natural selection. The same characteristics complicate genomic inference. The fecundity-with-high-mortality demography of *Illex illecebrosus* motivates the use of sweepstakes-reproductive-success models, wherein a small and variable fraction of adults contributes disproportionately to each generation (Hedgecock and Pudovkin, 2011; Eldon and Wakeley, 2006; Árnason et al., 2023). A large effective population size can simultaneously enhance the efficacy of weak selection, maintain abundant standing variation, and result in very short linkage disequilibrium, thereby complicating the disentanglement of population history and recent selection. More broadly, genomic analysis of cephalopods has lagged behind that of many commercially important vertebrates due to their large, repeat-rich, and extensively reorganized genomes relative to other molluscs (Albertin et al., 2015, 2022; Schmidbaur et al., 2022; Coffing et al., 2025, 2026). Until recently, these limitations were particularly pronounced for *Illex illecebrosus*, as no conspecific chromosome-level reference genome was available.

The inaugural comprehensive population genomic analysis of *Illex illecebrosus* established a crucial baseline. Utilizing whole-genome data from 540 specimens collected from over 40 locations, Baker et al. (2025) identified an absence of geographic genetic structure across the species’ Northwest Atlantic range. This finding aligns with the hypothesis of extensive connectivity facilitated by early-life dispersal and adult migration, thereby supporting the classification of *Illex illecebrosus* as a single, highly interconnected stock. Nevertheless, due to the lack of a conspecific genome assembly, the reads were aligned to the chromosome-level genome of the distantly related diamondback squid, *Thysanoteuthis rhombus*, which diverged approximately 51 million years ago, resulting in a mean alignment rate of merely ~46%. This divergence inherently limits inference to the conserved genome fraction that can be reliably mapped, thereby diminishing the ability to detect lineage-specific sequences, structural polymorphisms, and rapidly evolving loci. Consequently, Baker et al. did not find evidence of large structural variants, explicitly acknowledging reference divergence as a constraint on their detection capabilities.

Chromosomal inversions are particularly vulnerable to this issue because of their evolutionary significance, which arises not merely from their breakpoints but also from the extensive haplotypes they preserve. Inversions inhibit recombination between alternative configurations, thereby maintaining combinations of alleles as segregating “supergenes” (Kirkpatrick, 2010; Schwander et al., 2014; Thompson and Jiggins, 2014; Wellenreuther and Bernatchez, 2018; Faria et al., 2019). Numerous well-documented instances exhibit noticeable phenotypic effects or occur in spatially structured populations, such as mimicry in *Heliconius* (Jay et al., 2021), alternative reproductive morphs in the ruff (Lamichhaney et al., 2016), colony organization in fire ants (Wang et al., 2013), and life-history variation in *Mimulus* (Lowry and Willis, 2010). Consequently, much of the inversion theory has concentrated on local adaptation with migration, where suppressed recombination safeguards locally advantageous allele combinations from gene flow (Kirkpatrick and Barton, 2006; Charlesworth and

Barton, 2018). Inversions maintained at intermediate frequencies without geographic structuring present a distinct challenge. Their persistence suggests balancing mechanisms such as overdominance, associative overdominance induced by recessive deleterious load (Zhao and Charlesworth, 2016; Gilbert et al., 2020; Jay et al., 2021), negative frequency dependence, or temporally fluctuating selection (Wittmann et al., 2017; Rudman et al., 2022). Differentiating these histories from neutral drift necessitates reconstructing the age and frequency history of the arrangements while explicitly considering the recombination barrier between them (Navarro et al., 2000; Guerrero et al., 2012; Berdan et al., 2023).

A chromosome-level *Illex illecebrosus* genome facilitates the investigation of selection beyond large structural polymorphisms. Cephalopods present a unique context for this inquiry because of their extreme life history, which is associated with remarkable sensory and neural complexity. Coleoid genomes demonstrate significant lineage-specific reorganization (Schmidbaur et al., 2022), extensive RNA-editing-based transcriptomic plasticity (Liscovitch-Brauer et al., 2017), and sophisticated nervous and visual systems (Albertin et al., 2015, 2022; Styfals et al., 2022; Songco-Casey et al., 2022). In squids, vision adapts to substantial changes in depth and illumination during diel vertical migration, with photoreception extending beyond the camera eye to include dermal opsins and light-responsive chromatophore systems (Mäthger et al., 2010; Ramirez and Oakley, 2015; Kingston et al., 2015). In a large, highly connected population where geographic F_{ST} offers limited utility for detecting adaptation, recent positive selection is likely to manifest primarily through local distortions in diversity, linkage disequilibrium, and the site-frequency spectrum. Accurate interpretation of these signals necessitates first establishing the demographic and genealogical context in which they emerged.

In this study, we employed a chromosome-level annotated reference genome for *Illex illecebrosus*, recently made available (Small et al., 2026), to reanalyze range-wide whole-genome variation with significantly enhanced genomic resolution. Our dataset included 633 individuals distributed across 11 fishery management divisions (Baker et al., 2025), along with *I. coindetii* (Genoscope, 2025) as an outgroup for divergence-based inference and *I. argentinus* as a recently diverged congener (Small et al., 2026). Initially, we assessed whether the range-wide panmixia reported by Baker et al. (2025) persisted when reads were aligned to a conspecific reference and investigated the emergence of previously inaccessible structural variation. This analysis identified a single ~ 19 Mb inversion on chromosome 2 that segregated across the range, despite an otherwise largely unstructured genome. Subsequently, we determined whether this inversion represented a neutral polymorphism or a selected supergene by integrating arrangement-specific diversity and divergence with a barrier-aware coalescent model to reconstruct its age and historical frequency trajectory. Concurrently, we inferred population history from the site-frequency spectrum and linkage disequilibrium, explicitly evaluating whether the data necessitated a multiple-merger rather than Kingman coalescent, and scanned the remaining genome for recent positive selection using complementary methods (Kern and Schrider, 2018; Alachiotis and Pavlidis, 2018; DeGiorgio et al., 2016). These analyses elucidated two distinct forms of genomic adaptation in an otherwise panmictic species: a long-lived, balancing-selected chromosomal polymorphism and a genome-wide set of recent sweeps concentrated in neurodevelopmental and photoreceptive functions. Collectively, they demonstrated how a conspecific genome transformed the apparent genomic landscape of a highly connected marine species, revealing structural and adaptive variation that was largely undetectable through geographic differentiation alone.

2 Results

2.1 Sample, sex, and the absence of geographic structure

We conducted an analysis of 633 *Illex illecebrosus* individuals from eleven NAFO management divisions, covering approximately 37–50°N (fig. 1, table 1). Details of each collection event, including source and depth, are provided in Supplementary Table S8. The read coverage on the Z chromosome categorizes the sample into 342 ZZ males and 291 ZO females, aligning with a ZZ/ZO system (Coffing et al., 2025) (fig. 2). Autosomal analyses incorporate both sexes, while the Z chromosome, where males are diploid and females are haploid, is examined separately (section 2.5).

Consistent with the range-wide panmixia reported by Baker et al. (2025), population structure is observed in a specific genomic region, although it is not geographically linked. A windowed principal-component analysis along chromosome 2 identifies the three-cluster structure within the inversion (among-cluster variance $\eta^2 = 0.748$ within the inversion compared to $\eta^2 = 0.003$ in the collinear sequence, Supplementary Fig. S3), and two genome-wide tests confirm its uniqueness. First, the chromosome-2 karyotype does not account for variation on any other chromosome (in a per-chromosome PCA colored by karyotype, $\eta^2 = 0.76$ on chr2 versus ≤ 0.01 on all 45 others). Second, independent of the chr2 labels, a Gaussian-mixture screen of each chromosome’s leading principal component supports more than one cluster only on chromosome 2 (three clusters, $\Delta\text{BIC} = 199$, cluster separation 8.6σ) and on the Z chromosome (two clusters, reflecting the expected ZZ/ZO hemizygosity split), while every autosome forms a single cloud (fig. 2). The residual dispersion on minor PCA axes does not correlate with division, latitude, or coverage (Supplementary Fig. S1), and a KING-robust relatedness screen (Manichaikul et al., 2010) identifies no duplicate, parent-offspring, or full-sib pairs, and only a single second-degree pair among the 633 individuals (maximum kinship 0.16), indicating effective unrelatedness. Given that a per-SNP differentiation scan is the most sensitive test for local adaptation in such a panmictic population, we compared per-SNP F_{ST} among divisions to a coalescent neutral null, simulating a single panmictic population under the fitted demography at the empirical sample sizes and missingness. Across the complete per-chromosome callsets of all 633 individuals, thinned to every fiftieth SNP for tractability (551,344 autosomal SNPs, those in genes covering 12,007 genes), the empirical differentiation remains entirely within the drift expectation. No SNP surpasses the neutral maximum, only a single intergenic site exceeds the lenient threshold against approximately 556 expected by chance, and there are no genic outliers (Supplementary Figs. S2, S19). Nucleotide diversity and Tajima’s D are consistent across latitude, and pairwise F_{ST} among divisions is approximately 0 throughout (fig. 6). Consequently, there is no geographic local adaptation anywhere in the genome, and the inversion itself is geographically non-clinal (fig. 1, Supplementary Fig. S6 and Table S2). Furthermore, arrangement frequency does not correlate with capture depth, which ranges from 3 to 613 m across collection events (Supplementary Table S8). Among 539 individuals with recorded depth, the correlation between depth and the number of inverted chromosomes carried is $\rho = -0.08$ ($p = 0.08$), and within the six divisions sampled at multiple depths, it is $\rho = +0.04$ (within-division permutation $p = 0.49$). A weak negative slope across collection events ($p = 0.04$) is entirely attributable to the two shallow, jig-caught Newfoundland (3K) samples and disappears when they are excluded ($p = 0.23$), indicating it cannot be disentangled from gear and location effects.

2.2 A large, balancing-selected inversion on chromosome 2

Identity, breakpoints, and karyotypes. Local PCA (Li and Ralph, 2019) and a Gaussian-mixture karyotype assignment (95.3% confident) resolve three genotype clusters on chromosome 2 (254 AA, 284 AB, and 95 BB individuals, fig. 3a) whose differentiation is karyotypic rather than

geographic, with $F_{ST}(AA, BB) = 0.37$ inside the inversion against a geographic $F_{ST} \approx 0$. Between-karyotype F_{ST} computed along the whole chromosome is ≈ 0 (median 0.006) across collinear sequence and jumps to a median of 0.34 (maximum 0.53) inside the inverted body (Supplementary Fig. S20), so the karyotypes are one panmictic population everywhere except within the barrier. The three karyotypes co-occur throughout the range with no latitudinal cline in arrangement frequency (0.29–0.46 across divisions, fig. 1, Supplementary Fig. S6 and Table S2). Two independent lines pin the breakpoints to chr2:60.54–79.50 Mb (a ~ 18.96 Mb differentiated body), namely a step-change in arrangement differentiation and a strand flip in a whole-chromosome alignment to *I. coindetii* (Supplementary Fig. S11), and an extended block of linkage disequilibrium coincides with the same interval (Supplementary Fig. S4).

The reference genome carries the derived arrangement. The derived, inverted arrangement is BB, at a present frequency of 0.374, and AA is the ancestral, standard arrangement. This is the reverse of the naive polarisation from the reference allele, which is circular because the *Illex illecebrosus* reference assembly is itself in the inverted orientation (it carries BB at 86.5% of fixed-different sites). Restricting to the non-circular subset of sites at which *I. coindetii* carries the *Illex illecebrosus* alternate allele shows AA to be ancestral (63.0%) and BB derived. There was a single inversion event, and the reference was assembled from an inverted individual (Supplementary Table S3).

The inversion is non-neutral. The primary evidence is per-arrangement diversity (table 2, fig. 3c). Inside an inversion the two arrangements behave as partially isolated demes (Navarro et al., 2000; Guerrero et al., 2012), so at neutral equilibrium the diversity of each should scale with its frequency. The rare derived arrangement is instead more diverse than its frequency can sustain, with $\pi_I/\pi_S = 1.356 \pm 0.048$ against an equilibrium expectation of $p/(1-p) = 0.60$ for a polymorphism held at 0.374, a 2.3-fold excess that is the signature of an old polymorphism that was once common, not of a recent rise. Two diffusion arguments that use neither the mutation rate nor the accessibility mask exclude neutrality (section 5). The first concerns attainment. A neutral allele must first reach the inferred peak frequency of ~ 0.74 before it can decline, and the expected time for drift to carry a new mutation to that frequency is $2.1 N_e$ generations, which is ~ 1.15 million generations even at the smallest N_e on the trajectory (the ancestral 5.5×10^5), 1.36 times the entire inferred lifetime of the inversion, and up to 17 times at the present N_e . The probability that a new neutral inversion ever reaches 0.74 is of order 10^{-6} . The second concerns the decline. Over the fitted 100 ky fall, at the N_e prevailing then ($2.8\text{--}3.8 \times 10^6$), the standard deviation of allele frequency under pure drift is 0.055, so the observed fall of 0.366 is 6.7 standard deviations of drift. Even a fall four times slower would remain 2.5 standard deviations away, and a coalescent profile of the fall duration excludes the slow, drift-compatible alternative at $\Delta\chi^2 \gtrsim 8\text{--}80$ (Supplementary Fig. S14). Both arguments bound drift alone and take the fitted peak frequency as given, but together they establish that the arrangement was neither carried up nor brought down by drift.

Age and trajectory of the inversion. Estimating an inversion’s age requires a coalescent in which the two arrangements exchange ancestry only across the barrier, which standard simulators do not provide. We therefore fit the inversion’s age and frequency trajectory with msinv (Small, 2025), an ancestral-recombination-graph simulator that enforces the cross-karyotype recombination barrier, tracks ancestral material, and outputs tree sequences (section 5). The data constrain the shape of the frequency trajectory $p(t)$ rather than any site-level mechanism, so we condition on

a deterministic trajectory under within-arrangement neutrality and fit jointly to π_I/π_S , d_{xy}/π_I , and the per-arrangement singleton spectrum (fig. 4b, Supplementary Figs. S5 and S14). The coalescent age is ~ 850 ky. It is the robustly determined quantity because it is carried by d_{xy} between arrangements, which is symmetric to the polarisation. The reconstructed history is that the inversion arose ~ 850 ka, coincident with the onset of the population expansion (section 2.3), was maintained near $p \approx 0.74$ for most of its history, and then fell sharply to 0.374 over ~ 50 – 100 ky, ending $\gtrsim 50$ – 100 ka, under selection of order $N_e s \sim 10^2$ – 10^3 ($s \approx 1.5 \times 10^{-4}$). Simulated inversions of 37 kb rescaled to the 19-Mb body reproduce these statistics with length-invariance verified over a 25-fold range, and the fitted model reproduces the observed joint two-dimensional site-frequency spectrum between arrangements, with near-fixed differences and off-diagonal density in the inverted body and a collapse to the diagonal in collinear controls (Supplementary Fig. S5). Every absolute age scales with the assumed mutation rate $\mu = 3 \times 10^{-9}$.

Inversion as a balanced supergene. Recombination across the inversion is best read from linkage disequilibrium rather than from the recombination map. The map was estimated from all karyotypes together and from linkage disequilibrium within ~ 19 -kb windows, whereas the barrier acts only between arrangements and across the whole 19-Mb block. Within each window the many SNPs that segregate within arrangements dominate the signal, and the between-arrangement association carried by the few near-fixed differences is spread across megabases, so the map cannot resolve the barrier. We therefore report the map only for the collinear part of chromosome 2, which recombines at an ordinary autosomal rate (median 0.202 against an autosomal median of 0.200 cM/Mb, Supplementary Fig. S12). The barrier itself is shown directly by the block of linkage disequilibrium that spans exactly the inversion (Supplementary Fig. S4), by the arrangement differentiation, and by the joint site-frequency spectrum (fig. 3b, Supplementary Fig. S5). It is not locally adaptive, since there is no cline and geographic $F_{ST} \approx 0$, and the migration-selection scenario that explains most well-studied inversions (Kirkpatrick and Barton, 2006) cannot apply. And it is not a swept region, because the genome-wide scan below finds no cross-method-concordant sweep inside the inversion. The RAiSD footprint statistic is broadly elevated and the classifier over-calls across the inversion, but the two never co-localise into focal concordance, as expected for a balancing-selected supergene rather than a swept locus, and the concordance requirement filters these calls correctly. Gene flux between arrangements is not detectable in mean d_{xy} , though a residual $\sim 40\%$ depletion of near-fixed differences appears in the outermost megabase around the breakpoints. The mechanism of balancing selection, i.e., overdominance, associative overdominance, frequency dependence, or a moving optimum under fluctuating selection, is not resolved by these data. Though probable that a shelf-spawning squid whose paralarvae ride the Gulf Stream would experience temporally fluctuating selection on ~ 100 ky glacial cycles, however this remains a hypothesis.

Gene content and the architecture of arrangement divergence. The inversion has no functionally significant enrichment. Its 89 gene models (4.7 per Mb, against 6.8 per Mb genome-wide) place it at the 15th percentile of autosomal windows of the same size (median 127 genes, interquartile range 100–151), and the same holds for annotated content, since 45 genes have a cross-species functional annotation and 29 carry gene-ontology terms, again the 15th percentile (median 49 GO-annotated genes per window). Their functional classes are diverse (lipid metabolism, DNA repair, transcription and splicing, signalling) with no coherent pathway (Supplementary Table S7). A gene-ontology test of its annotated genes against the genome returns no enriched terms at $FDR < 0.05$, and a permutation null shows this is not a power artefact, because random gene sets of the same size produce more nominal hits than the inversion genes (86.6 ± 60 against 69, permutation

$p = 0.51$), while the largest nominal per-term enrichments (~ 40 -fold) rest on as few as two genes (Supplementary Fig. S21). The inversion has thus captured a functionally unremarkable slice of the genome, not a selected gene set but a supergene maintained for the arrangement as a whole. What the arrangements do accumulate is divergence as a unit, with 3,513 near-fixed differences between AA and BB (per-arrangement $|\Delta AF| \geq 0.9$) are spread across the inversion (Supplementary Fig. S5). These are overwhelmingly non-coding since coding sequence is only $\sim 0.2\%$ of the region, but the exonic fixed differences are ~ 2.3 -fold over-represented and clustered. All 19 fall in just nine genes, led by a cytochrome P450 (*CYP4X1*) and an acidic-repeat protein (four each) and including a phospholipase A2, the ciliary gene *RPGRIP1L*, and a cysteine-rich venom-like protein. Among coding sites inside the inversion, the most differentiated between arrangements are depleted rather than enriched for nonsynonymous (zero-fold degenerate) positions. Of the top 1% of coding sites by between-arrangement F_{ST} , 45% are zero-fold, against 82% of all coding sites in the body (odds ratio 0.18, $p = 2 \times 10^{-34}$), and per-transcript divergence between arrangements is unremarkable relative to collinear controls (5 of 95 transcripts exceed the control 95th percentile, as expected by chance, Supplementary Methods). Purifying selection therefore still removes deleterious differences between the arrangements, and any recessive load carried by the rarer arrangement is not yet visible as excess nonsynonymous divergence.

2.3 Demographic history shows an expanding population

The age and selection estimates above depend on the population’s demography and on the coalescent class being Kingman. Genome-wide, *Illex illecebrosus* shows $\pi \approx 0.0093$ and Tajima’s $D \approx -2$ in essentially every window (Tajima, 1989a,b), an unambiguous, genome-wide signature of expansion (table 3, fig. 4a, fig. 6). A moments fit to the folded SFS gives growth from an ancestral $N_e = 547,928$ to a present $N_e = 6,808,096$ over 769,519 generations (~ 12 -fold, with generations equal to years in this annual species), and this history reproduces both the observed π and a full-SFS Tajima’s $D = -1.92$ under forward simulation. Stairway Plot 2 and GONE2 concur on strong recent expansion to N_e in the $1.5\text{--}1.7 \times 10^7$ range. Because sweepstakes reproduction would mimic expansion by inflating rare variants (Eldon and Wakeley, 2006; Hedgecock and Pudovkin, 2011; Eldon et al., 2015), we tested the coalescent class directly. The folded SFS is fit far better by a Kingman-plus-growth model than by any multiple-merger (Beta) coalescent (Schweinsberg, 2003) (shape deviation $L_1 = 0.036$ vs a best- α 0.108), and contemporary N_e estimates from the SFS and from linkage disequilibrium agree where both are inconsistent with a strong multiple-merger process. The expansion began ~ 770 ka, i.e. at essentially the same time as the inversion arose, which places the supergene’s origin at the start of the demographic episode that shapes the whole genome.

2.4 Recombination, background selection, and protein adaptation

Three further features of the genome were resolved to accurately identify selective sweeps. Recombination was estimated with ReLERNN (Adrión et al., 2020), a deep-learning method that infers the rate from linkage disequilibrium. The estimated rate is ~ 0.20 cM/Mb on the autosomes (median 0.200, Supplementary Fig. S12). The map is flat with respect to gene content, and the rate varies by only $\sim 6\%$ between windows adjacent to exons and windows >100 kb from any exon, unchanged when π is held fixed (Supplementary Fig. S17a). For example, on chromosome 1 the rolling median spans just 0.17–0.23 cM/Mb, the pericentromeric flanks where π halves show no more than a 5% dip, and the centromere itself (a 2.5-Mb block that is 80% repeat and almost entirely masked) carries no windows (Supplementary Fig. S22). This pattern is similar genome-wide.

Across all 45 scanned autosomes the chromosome-scale rolling median varies by only 10–41% of each chromosome’s median, and no chromosome shows a peak more than 25% above it. Background selection is moderate and gene-density-driven. Nucleotide diversity falls with local coding density ($r = -0.31$, a 20–30% reduction in gene-dense windows) and rises 43% between exon-adjacent and exon-distant windows (Supplementary Fig. S17) (Charlesworth et al., 1993; McVicker et al., 2009; Charlesworth and Jensen, 2021). A model-based B -map could not be fit under this regime (the classical approximation overflows where the deleterious mutation rate exceeds the selection coefficient), so local coding fraction serves as the background-selection covariate that the sweep scan controls for explicitly. Protein adaptation over the divergence timescale is substantial. McDonald–Kreitman tests (McDonald and Kreitman, 1991; Messer and Petrov, 2013a) against *I. coindetii* across 20,999 genes give a genome-wide proportion of adaptive substitutions $\alpha \approx 0.34$ –0.64, with the splicing factor *SF3B4* the single most significantly adaptive gene (Supplementary Fig. S10). Per-gene power is limited, so we treat the MK results as complementary to the recent-sweep scan, which detects a different timescale of selection, recent and often non-coding. *I. argentinus* is a striking comparator. It is recently diverged (an order of magnitude fewer fixed coding differences than *I. coindetii*, Supplementary Table S3) yet carries near-identical genome-wide diversity ($\pi = 0.0092$ vs 0.0093, within 1%), so the two are large- N_e , expansion-shaped congeners of matched standing variation. We also found twenty-two blocks of ≥ 1 Mb on 17 chromosomes (129 Mb, 3.9% of the autosomes) that seemingly carry 2–6 times the chromosome-median π with less negative Tajima’s D (Supplementary Table S9). They are present at the same strength in every sampling location, are no more differentiated among locations than flanking sequence, and lack the long-range linkage disequilibrium of a segregating inversion (Supplementary Figs. S23, S24). Read-depth analyses resolve them into three categories. In six, depth is raised in every individual (1.2–2.2 times the flanks across the population sample, and up to 2.7 and 2.4 times in long reads from the individual used for the assembly and a second individual (Small et al., 2026)), and a single diploid genome carries five to six times the flanking density of heterozygous sites at skewed allele fractions. We interpret this as a duplicated sequence collapsed into one copy in the assembly. In eight, short-read depth is reduced to 0.6–0.9 times the flanks in every individual. A third of their accessible sequence is not uniquely mappable by 150-bp reads, against 4% genome-wide, so these blocks are largely duplicated within the assembly itself and short reads there map to several places and are lost at the mapping-quality filters. Long reads, which map uniquely through the duplicated sequence, show a much smaller deficit, and divergent haplotypes may add to the loss. The remaining eight have normal depth and mapping and are gene-poor regions of genuinely elevated diversity. Since our summary statistics cannot separate any of these from the footprints of selection, all 22 blocks were excluded from the sweep scan and the gene-set analyses.

2.5 A genome-wide scan for selective sweeps

We scanned for recent positive selection with diploS/HIC (Schrider and Kern, 2016; Kern and Schrider, 2018), a five-class site-frequency/diversity-based classifier (neutral, hard, soft, linked-hard, and linked-soft, in 1.1 Mb feature windows of eleven 100 kb sub-windows) trained on forward and coalescent simulations under the fitted growth demography with the final model reaching 81% class accuracy (fig. 5, Supplementary Figs. S7 and S8). Applied to the genome, the classifier over-calls sweeps, assigning only 1% of windows to the neutral class and 91% to the linked-hard or linked-soft classes. We determined this was because real neutral windows (accessible, ≥ 5 kb from any gene) carry 3–8-fold more within-window spatial heterogeneity in π and θ than the simulations, even though their marginal SFS matches ($D \approx -2$). We went further and simulated our identical missing-data structure and sample incompleteness ultimately ruling that out as the cause.

The residual is real per-sub-window diversity variation (mutation-rate variation plus background selection) that a uniform- θ coalescent does not reproduce (Johri et al., 2022). We therefore use the classifier as a stratified empirical-outlier detector, in which the sweep score ($p_{\text{hard}} + p_{\text{soft}}$) is ranked within strata of callable fraction and coding density, so a window is an outlier only relative to windows of similar accessibility and gene density, and calls are corroborated by an independent LD-footprint method, RAI_{SD} (Alachiotis and Pavlidis, 2018), or a composite-likelihood SFS method, SweepFinder2 (Nielsen et al., 2005; DeGiorgio et al., 2016). Thus, we report sweeps at two levels of stringency (Supplementary Fig. S13). Tier 1 is the diploS/HIC hard/soft classification, spatially resolved by a recombination-aware Markov decoder, a hidden Markov model over the per-window (100 kb) classes with neutral, linked, and sweep-centre states whose emissions are the prior-corrected class posteriors and whose dwell lengths follow the ~ 400 kb sweep footprint scaled by the recombination map (Supplementary Fig. S13a). Because the genome reads as $\sim 91\%$ linked-selected under strong expansion, the decoder’s role is spatial coherence. It retains hard/soft calls that lie in contiguous sweep centres and discards spatially isolated ones, reducing 1,651 raw calls to 677 windows (623 localised autosomal calls, 437 hard and 186 soft) and preferentially keeping the higher-outlier windows (median empirical-outlier percentile 0.98 vs 0.95 for discarded calls, Supplementary Fig. S13b). Tier 2 is the subset independently corroborated by a other methods, producing 24 concordant regions (a diploS/HIC-HMM outlier plus RAI_{SD} or SweepFinder2). The two tiers agree strongly, with 88% of windows at concordant loci decoded as sweep centres versus 2.9% genome-wide (odds ratio 242, $p < 10^{-30}$), and 21 of the 24 concordant regions contain a Tier-1 centre (Supplementary Fig. S13c), so the conservative concordance set is a high-purity subset of the classifier calls, not a disjoint list. The Z chromosome is necessarily diploS/HIC-only and scanned separately in males (below), it contributes four further Tier-1 calls, for a genome-wide Tier-1 total of 627 (fig. 5, Supplementary Fig. S18).

Neurodevelopment and photoreception. The 623 autosomal Tier-1 calls overlap 885 genes (475 with GO annotation). One-third of calls fall in gene-poor, often transposon-rich windows and are excluded from enrichment analyses. A gene-ontology enrichment against the annotated-gene background (Fisher’s exact test with Benjamini–Hochberg FDR, Benjamini and Hochberg 1995) returns 124 terms at FDR<0.05 (Supplementary Fig. S15, Table S4), dominated by neural development (neurogenesis, generation of neurons, neuron development, neuron-projection and cell morphogenesis, at 1.5–1.8-fold), together with transcriptional and RNA-regulatory control, and a weaker light-response signal (response to light stimulus 1.9-fold, response to radiation 1.8-fold). The enrichment is carried by the hard-sweep calls (98 terms), and the disjoint soft-sweep set has none at FDR<0.05. We further tested these regions against a hypothesis of background selection, since gene-dense regions have lower diversity for reasons unrelated to sweeps. We therefore screened every Tier-1 call with the same per-region background-selection test used for the concordant set, retaining only calls whose local π falls below the background-selection expectation given flanking coding density. This keeps 291 of the 623 (237 hard, 54 soft). The BGS-robust set yields 151 terms at FDR<0.05, and every leading term’s fold enrichment rises over the previous set, including response to light stimulus $1.94 \rightarrow 3.15$ (FDR $2 \times 10^{-2} \rightarrow 9 \times 10^{-5}$), response to radiation $1.76 \rightarrow 2.79$ (FDR $3 \times 10^{-2} \rightarrow 2 \times 10^{-4}$), neuron-projection morphogenesis $1.81 \rightarrow 2.25$, cell morphogenesis $1.73 \rightarrow 2.24$, neuron development $1.59 \rightarrow 1.97$, generation of neurons $1.59 \rightarrow 1.90$, transcription by RNA polymerase II $1.73 \rightarrow 2.08$ (Supplementary Fig. S15). The photoreception signal, the weakest in the full set, becomes the single most enriched theme after the BGS screen, and its more specific child terms (detection of light stimulus $2.5 \rightarrow 3.9$, phototransduction $2.5 \rightarrow 3.4$) move in the same direction, though with too few genes to reach significance. A shared-gene enrichment map of the

BGS-robust set resolves the same two connected modules, neural development/morphogenesis and light response, with no third (Supplementary Fig. S16, Table S5).

Candidate sweeps. The Tier-2 concordant regions carry a biologically coherent set of cephalopod-orthologous genes that sit squarely within these themes (table 4, Supplementary Table S6). They include the neural/EMT transcription factor *ZEB2*, the synaptic-vesicle GTPase *Rab3*, the apelin receptor *Aplnr* (cardiovascular/vasculature signalling), the DNA-damage-response gene *MACROD2*, the phospholipid synthase *cept1*, and the ceramide phosphoethanolamine synthase *Cpes*. A per-region background-selection test isolates 11 of these as high-confidence candidates in which diversity is reduced beyond the background-selection expectation, and fine-scale localisation confirms all 11 as recent sweeps by the sign of Tajima’s *D* at the diversity valley. It reassigns two apparent gene-desert signals to transposable-element regions (flagged lower confidence) and moves two targets to neighbouring genes, from *eif3h* to *Cpes* at chr1:85.5 Mb and to *GALT* at chr25:65.0 Mb. Both are genes already in the annotation, not new gene models. The single most distinctive autosomal candidate is a cephalopod-lineage-specific gene under a hard sweep on chromosome 16, a tandem-exon transcript with no functional annotation whose translation matches an uncharacterised protein family in the bobtail squid *Euprymna* ($E = 2 \times 10^{-53}$) (Albertin et al., 2015, 2022). There is no overlap between the sweep candidates and the MK-adaptive genes.

The Z chromosome. Because males are Z-diploid and females Z-haploid, the Z was scanned separately in the 330 confident ZZ males, using a model retrained at the matched sample size and accounting for the Z’s reduced effective size. The male-only screen was validated against the 289 haploid females, whose per-window diversity tracks the males’ at $r = 0.996$. Of the four Z Tier-1 calls (three hard and one soft, Supplementary Fig. S18), three credible, gene-localised sweeps survive fine-scale triage, all confirmed as conserved cephalopod genes by direct homology. These are *Pasilla*/Nova-1 (a neuronal alternative-splicing regulator with 92% identity to its *Octopus* ortholog), *Smaug*/samd4a (a germline RNA-binding translational repressor), and *Esrrg* (an estrogen-related receptor with an adjacent likely *cis*-regulatory lncRNA, with 97.5% identity to its *Octopus* ortholog), a coherent RNA-regulation-and-hormone-signalling signature on the sex chromosome. The *Amnionless*/“Zfest” sex-determination locus falls just outside the swept valley and shows only purifying selection, so it is a linked passenger rather than the target.

3 Discussion

The inversion is a datable, declining, balancing-selected supergene. Most inversions with understood evolutionary dynamics are maintained through local adaptation coupled with migration (Kirkpatrick and Barton, 2006; Lowry and Willis, 2010; Faria et al., 2019). However, the *Illex illecebrosus* inversion cannot be explained by this mechanism, as there is no geographical adaptation involved; the arrangement frequency is non-clinal, and geographic F_{ST} is consistently zero. Instead, it is maintained by balancing selection, with data providing precise constraints on its maintenance. The excess diversity of the rare derived arrangement suggests a historical prevalence of BB. Two mutation-rate-independent arguments and a barrier-aware coalescent model converge on an approximately 850,000-year-old polymorphism that plateaued near 0.74 and has been actively reduced to 0.37 over the past 50,000 to 100,000 years. This inversion differs qualitatively from classic *Drosophila* inversions, which follow clines and seasons (Corbett-Detig and Hartl, 2012; Kapun and Flatt, 2019), and from the human 17q21 inversion, which shows a recent increase under positive selection (Stefansson et al., 2005). It more closely resembles the mimicry supergene of

Heliconius numata, where an inverted arrangement carries a recessive deleterious load that induces associative overdominance, limiting its frequency (Jay et al., 2021). Associative overdominance is a plausible candidate here, as theory indicates that it arises in large, non-recombining blocks once recessive load accumulates (Zhao and Charlesworth, 2016; Gilbert et al., 2020; Berdan et al., 2021). The documented clustered coding divergence (19 exonic fixed differences in nine genes, within a block of 45 annotated genes lacking functional enrichment) provides the raw material for such a load, despite the differentiated coding sites not yet being enriched for nonsynonymous changes. However, a two-phase history of plateau followed by decline could also result from selection towards a shifting optimum (Matuszewski et al., 2015), and fluctuating selection is demonstrably strong in wild populations (Wittmann et al., 2017; Rudman et al., 2022). A shelf spawner with paralarvae advected by the Gulf Stream is a plausible target of 100,000-year climate cycles. The data constrain the trajectory but not the mechanism. They establish that the arrangement itself, rather than any single gene within it, is the unit of selection. The block is gene-poor, functionally unremarkable, differentiated as a unit, and free of sweeps, consistent with recombination suppression being the trait under selection (Kirkpatrick, 2010; Berdan et al., 2023).

Two methodological aspects of inversion analysis are broadly applicable. First, age determination relies on the polarization-symmetric divergence between arrangements, which remains intact even when the arrangement derivation is completely reversed. This reversal was necessitated by the discovery that the reference genome itself is inverted. Reference-orientation circularity poses a risk for any inversion study that polarizes from the reference allele, and outgroup-based polarization on non-reference sites serves as a solution. Second, the coalescent process within an inversion is characterized by a two-deme model with a barrier (Navarro et al., 2000; Guerrero et al., 2012). Accurate modeling requires a simulator that enforces this barrier within the ancestral recombination graph, rather than approximating the arrangements as freely recombining. The msinv tool provides this capability, and the plateau-and-decline history it uncovers is not accessible through methods that treat the inversion as a single locus. As the catalogue of wild inversions expands (Wellenreuther and Bernatchez, 2018; Mérot et al., 2020), the joint estimation of age, trajectory, and selection from per-arrangement diversity, divergence, and the joint site frequency spectrum (SFS) offers a means to classify them based on historical context rather than phenotype.

The sweep program is neural and photoreceptive, and it is robust to background selection. The genome-wide scan identified two primary themes. The first theme encompasses neurogenesis, neuron development, and neuron projection and cell morphogenesis, which aligns with expectations for a lineage characterized by a large, centralized, cell-type-rich nervous system (Albertin et al., 2015; Styfals et al., 2022). The genomes of this lineage exhibit regulatory reorganization around neural genes (Schmidbaur et al., 2022). A similar signature has been observed in a second coleoid lineage. In the sibling octopuses *Octopus bimaculatus* and *O. bimaculoides*, positive selection is concentrated in genes associated with brain and eye development (Coffing et al., 2026), suggesting that neural and visual genes may be recurrent targets of recent selection across cephalopods. The second theme, photoreception, is more unexpected and specific to *Illex illecebrosus*. Cephalopod photoreception extends beyond the camera eye, as opsins and phototransduction cascades are expressed in the skin and chromatophores (Ramirez and Oakley, 2015; Kingston et al., 2015). The visual system is highly diverse in cell types, with optic lobes developed convergently with those of vertebrates (Songco-Casey et al., 2022; Styfals et al., 2022; Yoshida et al., 2015). *Illex illecebrosus* is a fast, visually guided predator that migrates vertically each day through a steep light gradient and seasonally across latitude (O’Dor and Dawe, 1998), making selection on light detection biologically relevant. The decisive evidence that these themes are genuine, rather than

by-products of reduced diversity at gene-dense neural loci, is their behavior under the background-selection screen. Background selection reduces diversity where genes are dense (Charlesworth et al., 1993; McVicker et al., 2009; Charlesworth and Jensen, 2021), so a sweep call that is actually a BGS artifact is eliminated by a test requiring diversity below the BGS expectation, and any enrichment based on such calls should diminish. Instead, every leading term intensified (light response $1.9 \rightarrow 3.2$, radiation $1.8 \rightarrow 2.8$, morphogenesis $1.7\text{--}1.8 \rightarrow 2.2\text{--}2.3$), and the weakest theme became the strongest. The sweep program is not an artifact of gene density.

We present these as potential candidates rather than confirmed targets, and the framework itself is proposed as a contribution to this field. Simulation-trained per-window classifiers have become increasingly standard (Schrider and Kern, 2016; Kern and Schrider, 2018). However, on a rapidly expanding, unphased genome with abundant missing data, these classifiers tend to over-call because the actual neutral sequences exhibit greater spatial heterogeneity than manageable neutral simulations. Instead of concealing this issue behind a threshold, we quantified it and allowed it to guide our analysis. This was achieved through a stratified empirical outlier scan, cross-method concordance, and an explicit background-selection filter, aligning with recent recommendations for inference that models the confounders that it cannot eliminate (Johri et al., 2022). The predominance of hard sweeps over soft sweeps in the enriched set is noteworthy, as a population of this size is expected to adapt primarily through soft sweeps from standing variation (Messer and Petrov, 2013b). In contrast, an excess of hard sweeps suggests strong selection on new or rare variants with a limited mutational target (Hermisson and Pennings, 2005; Harris et al., 2018), in contrast to humans, where soft sweeps seem to predominate (Schrider and Kern, 2017). However, the soft-sweep class is underpowered in this context, and we refrain from drawing definitive conclusions regarding the mode of adaptation in this study.

Demography, sweepstakes, and licensing implications. A sweepstakes-reproduction species is precisely where multiple-merger coalescents are anticipated to mimic expansion (Eldon and Wakeley, 2006; Hedgecock and Pudovkin, 2011). Recent research on Atlantic cod suggests that pervasive recurrent sweeps can independently generate multiple-merger genealogies (Árnason et al., 2023). We directly tested the coalescent class rather than presuming it, revealing that the site frequency spectrum (SFS) aligns with Kingman-plus-growth rather than Beta, and that SFS- and linkage disequilibrium (LD)-based effective population size (N_e) are consistent. This finding underpins all age and selection estimates presented, indicating that *Ilex illecebrosus*’s substantial fecundity has not, at the genomic scale we can assess, resulted in the skewed genealogies its life history might imply. The emergence of the inversion coinciding with the onset of expansion is noteworthy, as a supergene established at the beginning of a ~ 12 -fold demographic expansion would have maintained its plateau frequency throughout the expansion, with its subsequent decline being a recent occurrence in this context.

4 Limitations

The callset remains unphased, precluding the availability of haplotype-homozygosity statistics. Consequently, sweep detection is reliant solely on the site frequency spectrum (SFS) and diversity, which particularly limits its efficacy against soft sweeps. The mismatch between simulation and real neutral scenarios, which leads to classifier over-calling, persists unresolved at the level of absolute neutral calling. Our empirical-outlier framework serves as a principled response rather than a solution, and it also negates a within-arrangement SFS approach to inversion analysis. The age of each inversion is inversely proportional to the mutation rate, with $\mu = 3 \times 10^{-9}$ being

the least robust external input. An independent calibration, such as using a karyotyped congener like *I. argentinus*, offers the most definitive path to determining age without reliance on mutation rates. The inversion model is predicated on a deterministic frequency trajectory and assumes within-arrangement neutrality, thereby constraining the form of $p(t)$ without addressing the mechanism of balancing selection. The linkage disequilibrium (LD)-based recombination map exhibits a compressed dynamic range at the chromosomal scale and fails to resolve centromeric suppression (Supplementary Fig. S22). It is employed solely for its genome-wide mean and for scaling the decoder’s dwell time, and the absence of a diversity-recombination relationship in these data should be interpreted with this limitation in mind. The corroborating footprint methods are broad, and the recombination-aware decoder is informed by design rather than calibrated through simulation. Candidate sweep genes are identified based on homology to distant cephalopod models and cross-method concordance, with the two enrichment themes reflecting statements about gene sets rather than individual confirmed targets.

5 Methods

The main steps are summarised here. Parameter values, thresholds, and software versions are given in the Supplementary Methods and Supplementary Table S1. The scripts for every step are in the `pipeline/` directory of the manuscript repository, and the inversion analysis is in the `msinv` repository (Data and code availability).

Samples and variant calling. We analysed all 633 *Illex illecebrosus* individuals whole-genome sequenced by Baker et al. (2025) (ENA project PRJEB101095), collected in 2022 and 2023 across eleven NAFO divisions, with coordinates, date, and capture depth recorded for each collection event (Supplementary Table S8). That study analysed 540 of them after removing individuals with more than 25% missing genotypes, a consequence of mapping to the distantly related *Thysanoteuthis* reference. We re-processed the raw reads from scratch against the conspecific `Illex_F24` primary assembly (45 numbered chromosomes plus the Z), and against this reference no individual needed to be excluded. Reads were trimmed with `fastp` (Chen et al., 2018), mapped with `BWA-MEM2` (Vasimuddin et al., 2019), and deduplicated with `Picard`, and genotypes were called jointly across individuals with `GATK HaplotypeCaller` in `gVCF` mode (Poplin et al., 2018), all within `gengenpipe` (Czech and Exposito-Alonso, 2022). The joint callset retains invariant sites. Variants were filtered on site quality, mapping quality, strand bias, missingness, and excess heterozygosity at intermediate frequency, and sites in GC-corrected high-depth windows or in annotated repeats were excluded, leaving 1.43 Gb of accessible sequence, of which 96% is uniquely mappable by 150-bp reads according to `GenMap` (Pockrandt et al., 2020). For the sweep scan and gene-set analyses this accessibility mask was extended by the 22 anomalous-diversity blocks described in the Results (Supplementary Table S9). Genotypes are unphased.

Diversity, site-frequency spectra, and demography. Mean sequencing depth is about 9-fold, so diversity statistics were estimated from genotype likelihoods rather than from called genotypes. Folded site-frequency spectra, π , Watterson’s θ , and Tajima’s D were computed with `ANGSD` (Korneliussen et al., 2014) directly from the BAM alignments rather than from the VCF, genome-wide, per chromosome, per NAFO division, and per karyotype, with the Z analysed in males only. Demographic history was inferred with `Stairway Plot 2` (Liu and Fu, 2020), moments (Jouganous et al., 2017), and `GONE2` (Santiago et al., 2025), and the fitted model was validated by forward simulation. The coalescent class was tested by comparing the folded spectrum with Kingman-plus-growth

and Beta-coalescent expectations. Statistics that require individual genotypes were computed on the called genotypes with `pg_gpu` (kr-colab, 2025). These are the windowed per-arrangement π and d_{xy} , the between-karyotype and between-division Hudson F_{ST} (Bhatia et al., 2013), linkage disequilibrium, and the principal-component analyses, with missing genotypes retained and handled pairwise. Per-SNP geographic F_{ST} was computed with `plink2` (Chang et al., 2015) using the Weir–Cockerham estimator. Relatedness was screened with `KING-robust` (Manichaikul et al., 2010).

The chromosome-2 inversion. Karyotypes were assigned by local PCA (Li and Ralph, 2019) and a Gaussian-mixture model on the first principal component. The arrangements were polarised from an `AnchorWave` (Song et al., 2022) whole-genome alignment to *I. coindettii*, using only sites at which the outgroup does not simply match the reference, with `est-sfs` (Keightley and Jackson, 2018) as a check. The age and frequency trajectory were fitted with `msinv` (Small, 2025), an ancestral-recombination-graph simulator that enforces the recombination barrier between arrangements (Supplementary Methods). Gene content, fixed differences between arrangements, gene-ontology enrichment against a permutation null, the drift bounds, and the mutational-load test are described in the Supplementary Methods.

Recombination. A recombination map was estimated with `ReLERNN` (Adrion et al., 2020), which trains a recurrent neural network on coalescent simulations matched to the fitted demography and the accessibility mask and then predicts the rate from linkage disequilibrium in windows of about 19 kb. The map was built from the 329 confidently sexed males that had calls on every chromosome, one of 330 being removed, and a bootstrap correction was applied.

Selective sweeps. We classified 100-kb windows with `diploS/HIC` (Kern and Schrider, 2018), trained on `SLiM` forward simulations (Haller et al., 2026) recapitated with `msprime` (Kelleher et al., 2018; Haller et al., 2019) and on neutral `msprime` simulations under the fitted demography, at the empirical sample size and with the empirical pattern of missing data. Training and scanning used the same unphased feature extractor and the full site-frequency spectrum. Each window’s sweep score was ranked within strata of callable fraction and coding density, and outliers were corroborated with `RAiSD` (Alachiotis and Pavlidis, 2018) and `SweepFinder2` (DeGiorgio et al., 2016). A recombination-aware hidden Markov model resolved the window classifications into spatially coherent Tier-1 calls. The Z was scanned in males only with a model retrained at their sample size. The extended mask excluded the 22 anomalous-diversity blocks throughout. Candidate genes were annotated with `EnTAP` (Hart et al., 2020) and checked against cephalopod proteomes with `DIAMOND` (Buchfink et al., 2021). Gene-ontology enrichment used Fisher’s exact test with Benjamini–Hochberg correction (Benjamini and Hochberg, 1995), and the background-selection screen retained calls whose local diversity lies at least 0.003 below the value expected from coding density within 1 Mb (Supplementary Methods).

McDonald–Kreitman tests. McDonald–Kreitman tests used `mkado` (Rivera-Colón et al., 2026) with *I. coindettii* as the outgroup for 20,999 genes, with codon degeneracy annotated by `degenotate` (gwct, 2024).

Table 1: **Sample and whole-genome summary.** Diversity statistics are genome-wide from genotype likelihoods, and karyotype counts and the inverted-arrangement frequency are for chromosome 2.

Quantity	Value
Species / callset	<i>Illex illecebrosus</i> ; Baker et al. (2025) whole-genome
Individuals (karyotyped)	633
Sex (chrZ coverage)	342 ZZ males / 291 ZO females
Sampling	11 NAFO divisions, ~37–50°N
Geographic structure	$F_{ST} \approx 0$ (none outside chr2)
Genome-wide π	0.0093
Genome-wide Tajima’s D	−2.07 (−2.09 in 50 kb windows)
chr2 karyotypes (AA/AB/BB)	254 / 284 / 95
Inverted (BB) frequency	0.374 (0.29–0.46 across divisions, non-clinal)
Karyotype F_{ST} (AA,BB)	0.365

Data and code availability

The manuscript, along with all figures and tables, their generating scripts, and editable table exports, is available in the manuscript repository (https://github.com/stsmall/Illex_popgen). A section-by-section index of every figure and table source file is provided in FIGURES_AND_TABLES_INDEX.md. The annotation utilized is `Illex_F24` version 3, which includes additions to version 2, specifically the loci recovered in the inversion and the chromosome-16 functional note, located in `data/annotation/` within the same repository. The coalescent simulations for the inversion analysis (section 2.2) were generated using `msinv` v0.3.5, a general ARG-based inversion-coalescent simulator (<https://github.com/stsmall/msinv>, MIT, commit `abd0d63`). The complete *Illex illecebrosus*-specific inversion analysis, including fitting code, simulation configuration, tabulated outputs, and inversion figures, is contained within the repository under `illex/` and `results/illex/`.

Acknowledgements

[TODO: funding, contribution?.]

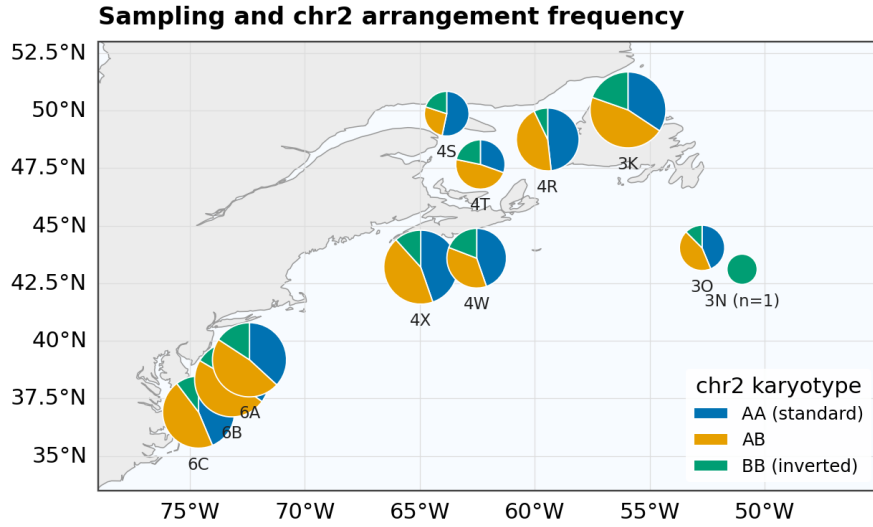


Figure 1: **Sampling and chromosome-2 arrangement frequency.** *Illex illecebrosus* was sampled across eleven NAFO divisions (approximately 37–50°N). The pie charts, positioned at the mean sampling coordinates of each division, illustrate the AA/AB/BB karyotype composition, with sizes proportional to N . The frequency of the inverted (BB) arrangement is non-clinal (ranging from 0.29 to 0.46 across latitudes), which aligns with a geographic $F_{ST} \approx 0$, indicating that the inversion is not locally adaptive.

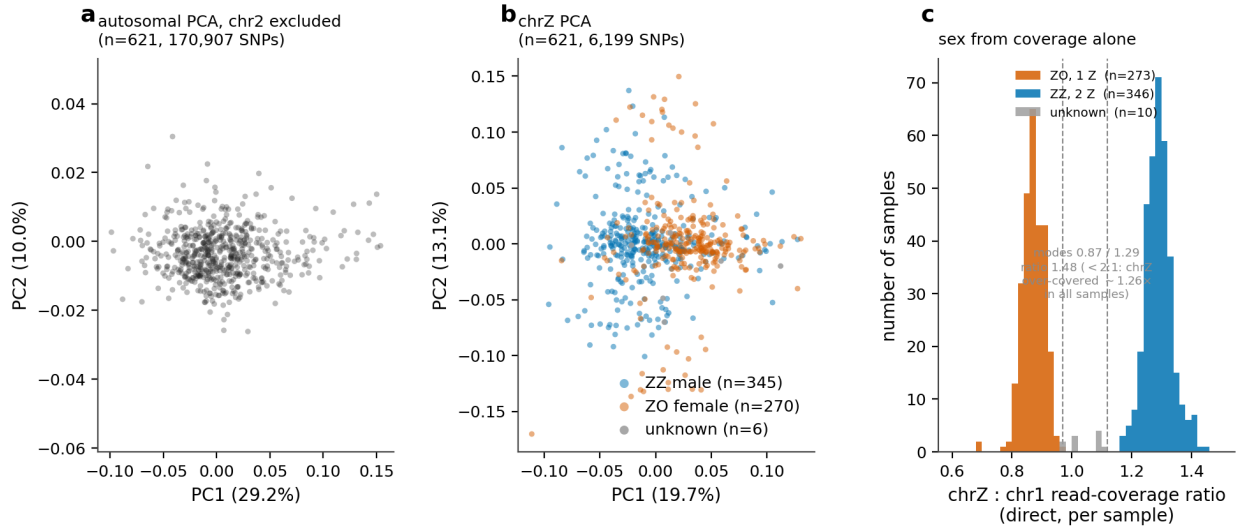


Figure 2: **Absence of population structure outside chromosome 2 and sex assignment** (consistent with the range-wide panmixia reported by Baker et al. 2025). (a) Genome-wide autosomal PCA excluding chromosome 2 reveals a single undifferentiated cluster. (b) PCA of chrZ shows mild separation between males and females. (c) The chrZ:chr1 read-coverage ratio per individual indicates ZZ males at approximately 1.0 (two Z copies), ZO females near the single-copy expectation (absence of W), with individuals in the ambiguous band remaining unassigned. The sample includes 342 males and 291 females.

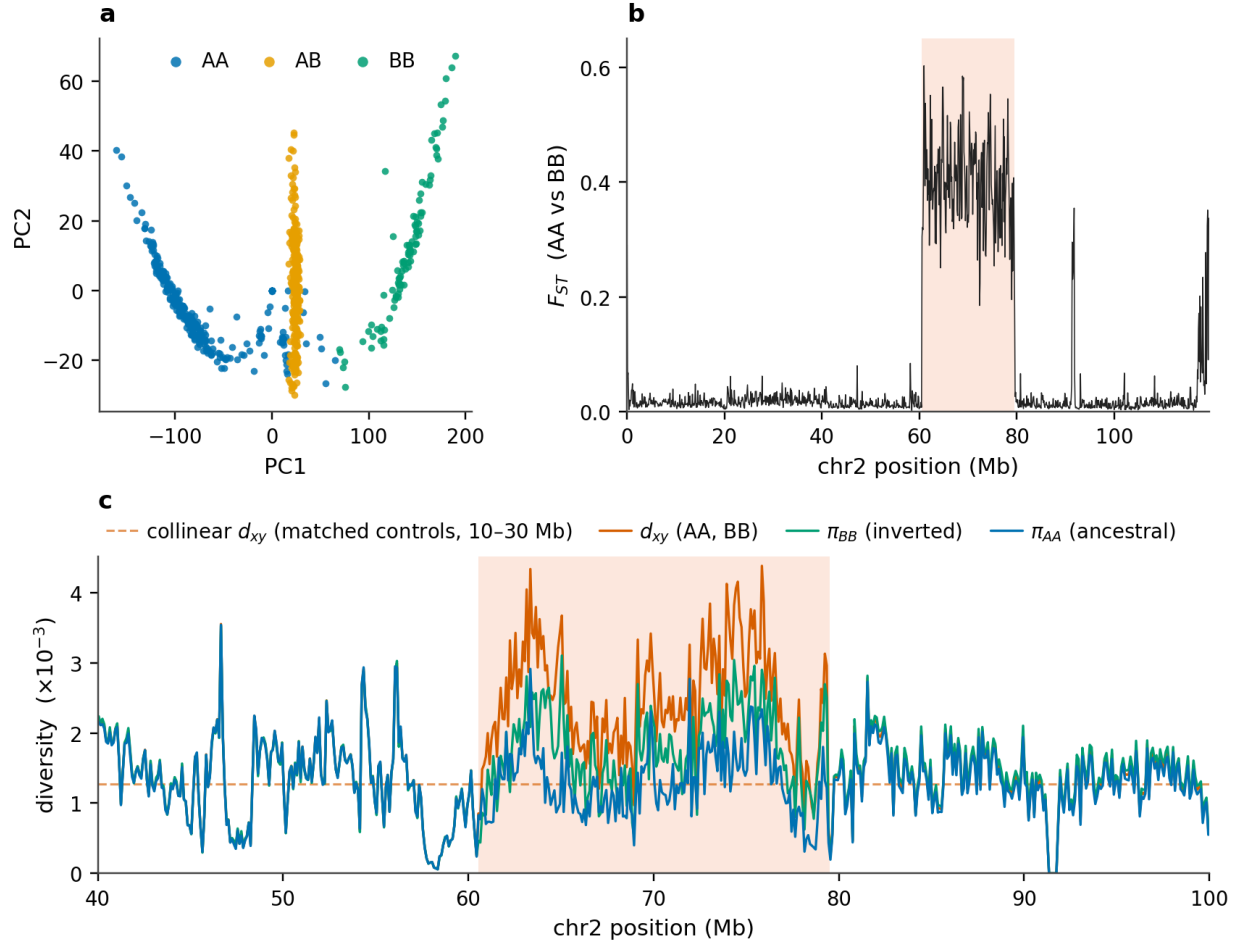


Figure 3: **Chromosome-2 inversion analysis.** (a) PCA of chr2 distinguishes three karyotype clusters (AA/AB/BB). (b) F_{ST} values between arrangements along chr2 indicate that the inversion (60.54–79.50 Mb, shaded) is the sole differentiated region (median $F_{ST} = 0.37$ compared to 0.01 in collinear regions, with a comprehensive view in Supplementary Fig. S20). (c) Diversity per arrangement across the inversion (shaded) and 20 Mb of collinear sequence on either side shows that within the breakpoints, the derived, inverted BB arrangement is more diverse than the ancestral AA ($\pi_I/\pi_S = 1.36$), despite its lower frequency, and d_{xy} surpasses both, indicating an ancient, previously common polymorphism. Outside the breakpoints, the three curves converge at the collinear d_{xy} (dashed), as expected where arrangements recombine freely.

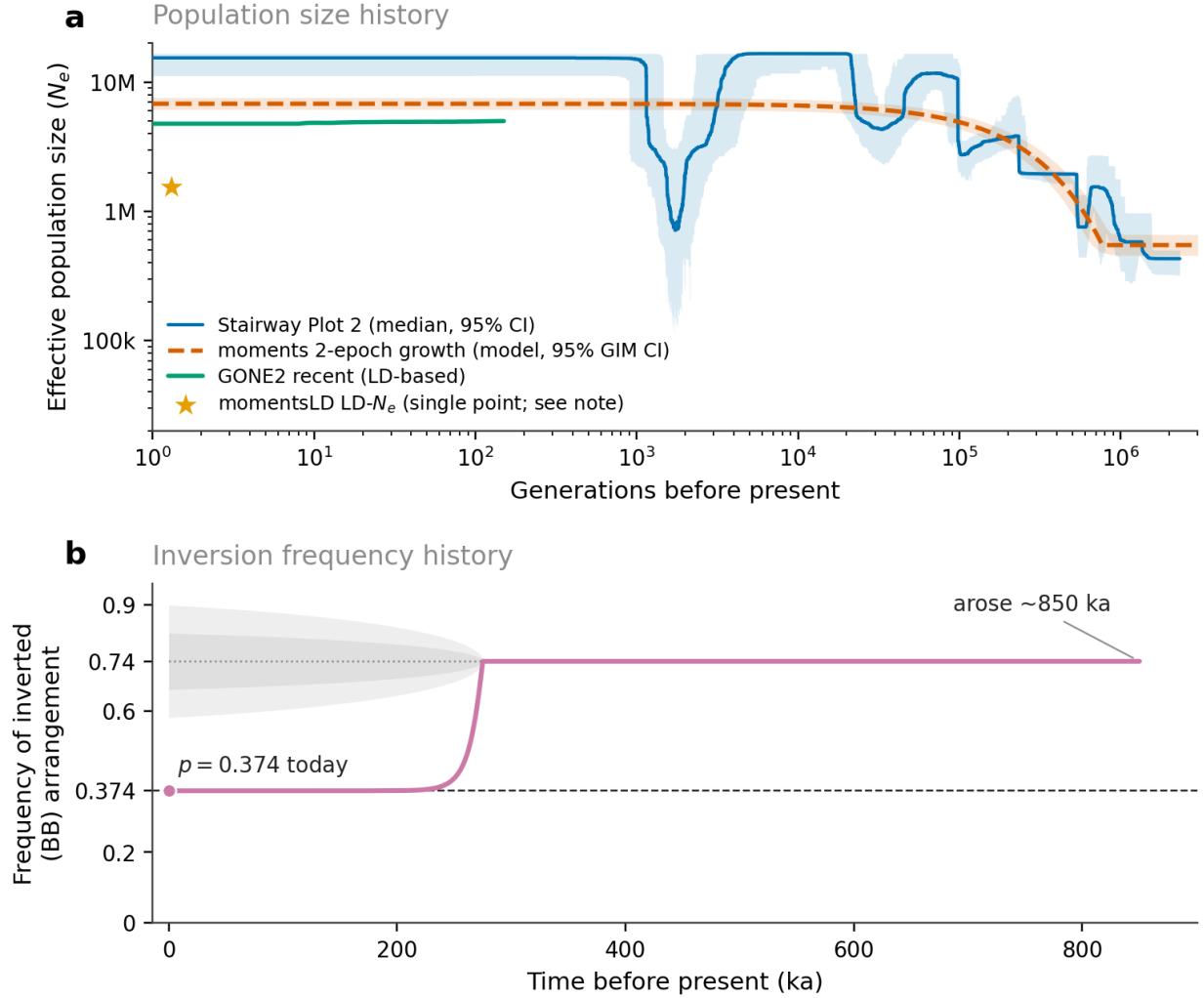


Figure 4: Demographic history and inversion frequency history. (a) The effective population size over time is depicted using Stairway Plot 2 (median $\pm 95\%$ CI), alongside the moments 2-epoch growth model with its Godambe (GIM) 95% parameter-CI envelope (expansion $12.4 \times \pm 9\%$, $T \approx 770$ ky $\pm 16\%$). GONE2 is utilized for recent historical data, and a single momentsLD LD- N_e point is presented (rate-corrected, with only the contemporary point shown due to the non-identifiability of its time-varying fit). (b) The inversion trajectory is fitted using the barrier-aware coalescent simulator msinv. The derived (BB) arrangement emerged approximately 850 ka, coinciding with the onset of expansion in (a), maintained near $p \approx 0.74$, and subsequently declined to 0.374. The neutral-drift envelope is provided for comparison. $\Delta\chi^2$ profiles, which identify the end and duration of the decline, are available in Supplementary Fig. S14, indicating that the decline was recent and rapid, excluding neutral drift-out.

Table 2: **The chromosome-2 inversion** (corrected polarisation, with I = BB inverted/derived and S = AA ancestral/standard).

Quantity	Value
Location (differentiated body)	chr2:60.54–79.50 Mb (~ 18.96 Mb)
Karyotypes (AA/AB/BB)	254 / 284 / 95 (95.3% confident)
Inverted (BB) frequency p	0.374
Historical peak frequency	~ 0.74
Karyotype F_{ST} (AA,BB)	0.365 (geographic $F_{ST} \approx 0$)
π_I/π_S (I=BB)	1.356 ± 0.048
d_{xy}/π_I	1.385 ± 0.021
Coalescent age t_{inv}	~ 850 ky ($\propto 1/\mu$, $\mu = 3 \times 10^{-9}$)
Decline	$0.74 \rightarrow 0.374$ over ~ 50 – 100 ky, ending $\gtrsim 50$ – 100 ka
Selection on decline	$N_e s \sim 10^2$ – 10^3 ($s \approx 1.5 \times 10^{-4}$)
Gene content	89 gene models, 45 annotated; no GO enrichment ($p = 0.51$)
Fixed differences (AA vs BB)	3,513; 19 exonic, in 9 genes

Table 3: **Demographic model** (moments, folded SFS, validated by forward simulation).

Parameter	Value
Ancestral N_e (N_REF)	547,928
Present N_e (N0)	6,808,096
Growth duration (T_GROW)	769,519 generations
Fold expansion	$\sim 12\times$
Genome-wide π	0.0093
Genome-wide Tajima’s D	≈ -2 (nearly all windows)
Mutation rate μ	3×10^{-9} / bp / generation
Recombination (autosomal)	~ 0.21 – 0.24 cM/Mb (sex-averaged)

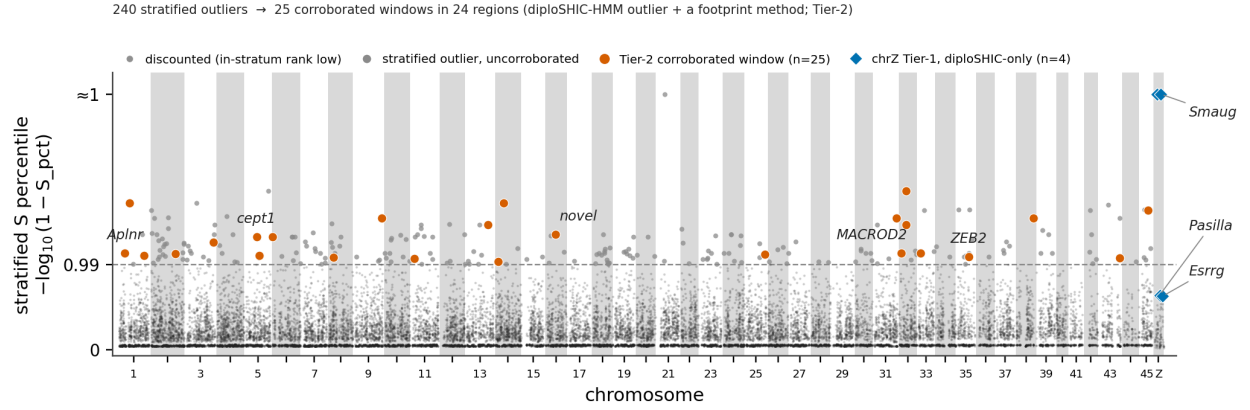


Figure 5: Genome-wide selective-sweep scan. The sweep signal across the genome (chromosomes concatenated) is illustrated. The classifier interprets the genome as predominantly linked-selected ($\sim 91\%$ linked-hard or linked-soft, $\sim 1\%$ neutral, and only a few percent hard or soft). Raw diploS/HIC calls are filtered by cross-method concordance (a diploS/HIC-HMM outlier plus RAI_{SD} or SweepFinder2) and a background-selection test. The corroborated candidates, robust against background selection (25 windows across 24 regions, highlighted), are distinguished above the discounted background, with key genes labeled. Chromosome-Z Tier-1 calls (diamonds, with *Pasilla*, *Smaug*, *Esrrg* labeled) are male-based and diploS/HIC-only (the footprint methods do not cover the Z), thus they cannot be included in the concordance filter. The chromosome-2 inversion does not exhibit a concordant sweep. The complete scan pipeline is detailed in Supplementary Fig. S8.

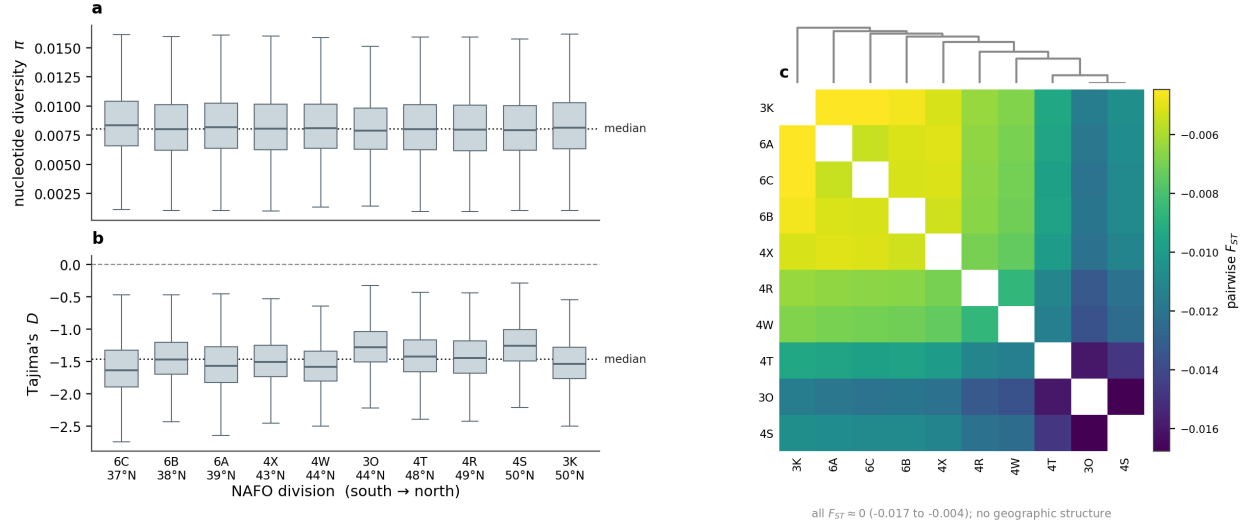


Figure 6: **Diversity is uniform and there is no geographic structure.** (a) Nucleotide diversity π and (b) Tajima's D per NAFO division (ten of the eleven sampled divisions, as 3N, with a single individual, cannot support per-population estimates), ordered from south to north, and both are flat across latitude (no cline), and D is negative in every population, the signature of strong expansion. (c) Pairwise F_{ST} among divisions as a clustered heatmap with dendrogram. Every value is ≈ 0 (-0.017 to -0.004) and the clustering follows sample size, not geography, i.e. no population structure. (π/D estimated by ANGSD genotype likelihoods over all 45 autosomes plus the Z, large divisions subsampled to $n = 25$, and F_{ST} genome-wide by pg_gpu. Per-chromosome distributions are in Supplementary Fig. S9.)

Table 4: **High-confidence selective-sweep candidates**, grouped by biological function and ordered by chromosome, including the Z-chromosome set. Autosomal rows are the 11 background-selection-robust candidates (of 24 concordant regions), and Z rows (marked) are the fine-scale-confirmed sweeps from the males-only scan. Genes are cephalopod orthologs. Class is hard or soft sweep. Full machine-readable tables are in Supplementary Table S6.

Location	Gene / feature	Class
<i>Neural, RNA-regulation & development</i>		
chr35:39.0 Mb	<i>ZEB2</i> (neural/EMT transcription factor)	hard
Z:20.8 Mb	<i>Pasilla</i> /Nova-1 (splicing regulator)	hard (Z)
Z:23.6 Mb	<i>Smaug</i> /samd4a (translational repressor)	hard (Z)
<i>Cardiovascular</i>		
chr1:20.3 Mb	<i>Aplnr</i> (apelin receptor)	hard
<i>Genome-maintenance & stress</i>		
chr32:9.3 Mb	<i>MACROD2</i> (DNA-damage response)	soft
<i>Membrane & lipid</i>		
chr5:38.0 Mb	<i>cept1</i> (phospholipid synthesis)	hard
chr1:85.5 Mb	<i>Cpes</i> (ceramide phosphoethanolamine synthase) [†]	hard
<i>Metabolism</i>		
chr25:65.0 Mb	<i>GALT</i> [†]	soft
<i>ECM & adhesion</i>		
chr33:9.0 Mb	<i>Tinagl1</i>	hard
<i>Signalling</i>		
chr32:24.1 Mb	GPCR cluster	hard
Z:30.7 Mb	<i>Esrrg</i> (estrogen-related receptor)	soft (Z)
<i>Cephalopod-specific novelty</i>		
chr16:35.3 Mb	uncharacterised (<i>Euprymna</i> family)	hard
<i>Lower confidence (TE-dense)</i>		
chr1:35.9 Mb	transposable element [‡]	hard
chr11:10.8 Mb	transposable element [‡]	hard

[†] target reassigned by fine-scale localisation to a neighbouring annotated gene. [‡] TE-dense region, where deep π and negative D may be a TE-insertion sweep or a repeat artifact.

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